Comparing Measures of Centrality in Bipartite Social Networks: A Study of Drug Seeking for Opioid Analgesics

Brian Aronson[a], Kai-Cheng Yang[b], Meltem Odabas[a], Yong-Yeol Ahn[b,c], Brea L. Perry[a,c]

Abstract

Social network analysts typically study node prominence in bipartite social networks with methods that do not fully account the two-mode topology of the underlying social network. Recent studies have generated a number of algorithms that might estimate node prominence in bipartite graphs with greater accuracy; however, it is unclear how these algorithms perform on real world social networks. To provide a comparison of bipartite centrality algorithms and recommendations regarding their appropriate use cases, this article examines their performance in a real-world network of patient-provider ties. We build on a recent body of research showing that network centrality in patient-provider networks is a promising indicator of prescription drug-seeking. We measure centrality with recently-developed bipartite methods, and we find that BiRank and CoHITS provide significantly more robust measures of prescription drug-seeking and better predictors of subsequent opioid overdose than traditional centrality estimates, including PageRank in a one mode network projection. Our results suggest that these algorithms may be very useful for identifying centrality across network contexts.

Affiliations

a) Department of Sociology, Indiana University Bloomington, Ballantine Hall, 1020 E Kirkwood Ave, Bloomington, IN 47405, USA
b) School of Informatics, Computing, and Engineering, Indiana University, Bloomington, IN 47408, USA
c) Network Science Institute, Indiana University, 1001 45/46 Bypass, Bloomington, IN 47408, USA

Email addresses

Brian Aronson: briarons@iu.edu
Kai-Cheng Yang: yangkc@iu.edu
Brea L. Perry: blperry@indiana.edu
Yong-Yeol Ahn: yyahn@iu.edu
Meltem Odabas: modabas@iu.edu

1. Introduction

Centrality is a key structural feature of social networks, communicating important information about an individual's prominence or role within a given network (Borgatti & Everett 1997; Hanneman & Riddle 2005). For example, a scientist's centrality in a scientist-paper co-citation network typically corresponds to the scientist's degree of interdisciplinary bridging and opinion leadership (Moody & Light 2006; Otte & Rousseau 2002); a patient's centrality in a patient-provider prescription network may reflect the extent to which the patient is deliberately searching for lax or complicit opioid prescribers (Perry et al. 2015, 2019). Both scientist-paper and patient-provider networks are examples of bipartite (or two-mode) networks, where social ties only occur across but not within two distinct sets (or modes) of actors (Borgatti & Everett 1997; Wasserman et al. 1994). In the case of scientist-paper co-citation networks, scientists are only affiliated through their shared citations; in patient-provider prescription networks, patients are only affiliated to each other through their shared providers.

Despite substantial research interest in measuring node prominence, many algorithms, such as PageRank or eigenvector centrality, are designed primarily for unipartite (one-mode) networks. In these cases, social ties are assumed to be unconstrained and possible between every pair of nodes (Bonacich 1991). These centrality algorithms cannot take into account that ties across each mode of a bipartite social network are never present; therefore, when applied to bipartite social networks, these centrality algorithms can provide misleading results (Latapy et al. 2008; Melamed 2014). Indeed, the biases induced by applying eigenvector-based centrality algorithms to bipartite social networks have been a source of concern in social network theory for decades (Bonacich 1991; Borgatti & Everett 1997; Everett 2016; Zhou et al. 2007).

More recently, researchers in information science have developed centrality algorithms for digital networks that can better take into account the topology of bipartite graphs; however, these methods have rarely been applied in social network analysis (Deng et al. 2009; He et al. 2017; Rui et al. 2007; Zhou et al. 2007). Bipartite ranking algorithms such as HITS, CoHITS, BGRM, and BiRank operate similarly to eigenvector-based methods in that they iteratively update node centrality estimates based on each node's connectivity and walk distance to other prominent nodes in the social network; however, bipartite algorithms differ in that they update node ranks based on the structure of nodes within both modes of the graph simultaneously. Specifically, across each iteration of bipartite ranking algorithms, ranks for each node are updated based on each node's walk distance to other nodes in the graph, each node's estimated centrality during the prior iteration of the bipartite algorithm, and the most recent rank estimates of each node's connection on the other mode of the graph.

Because bipartite centrality algorithms account for the topology of bipartite network structures, they are likely to produce more accurate estimates of node prominence than centrality estimates derived from traditional eigenvector-based methods. Nonetheless, three gaps in the literature have limited the adoption of bipartite centrality algorithms in studies of social networks: 1) Because bipartite algorithms were primarily developed for estimating node prominence in digital networks, it remains unclear how well bipartite centrality algorithms actually generalize to social networks; 2) There are a number of bipartite ranking algorithms now available, but there remains limited guidance about which bipartite ranking algorithm is most appropriate for estimating centrality in social networks; 3) Until recently, developers of bipartite centrality algorithms offered little information about how to implement bipartite centrality measures in popular statistical programs (Aronson & Yang 2020; Yang 2020; Yang et al. 2020).

Filling these gaps is critical for importing these new bipartite centrality algorithms into social network analysis.

To test the applicability of bipartite centrality algorithms in social networks, this paper studies the output from four bipartite centrality algorithms (HITS, CoHITS, BGRM, and BiRank) (Deng et al. 2009; He et al. 2017; Kleinberg 1999; Rui et al. 2007) and one unipartite centrality algorithm (PageRank) (Page et al. 1999) in a real-world social network of patient–provider relationships. Guided by prior research, we track how well different centrality measures map onto explicit drug seeking behavior (also known as "doctor shopping") among patients, and then use them to predict subsequent opioid overdose. We also propose a novel method for weighting bipartite centrality measures by node-attributes, and illustrate how this method can improve our ability to capture specific aspects of prescription drug-seeking that centrality on the unweighted network cannot. Overall, our study both builds on methods research about estimating centrality in bipartite social networks (Everett 2016; He et al. 2017) and contributes to a growing literature about how social network analysis can be used to improve analysts' ability to identify fraud and abuse using insurance claims data or electronic health records (Ong et al. 2016; Perry et al. 2019; Stein et al. 2017; Takahashi et al. 2016).

2. Prescription Opioid-Seeking for Nonmedical Use

Visiting multiple health care providers is a widespread method for obtaining opioids for nonmedical use, and likely played an important role in fueling the United States opioid epidemic (Maxwell 2011; Peirce et al. 2012). From 1999 to 2018, age-adjusted rates of mortality by opioid overdose have increased by approximately 240%, from 6.1 per 100,000 in 1999 to 20.7 per 100,000 in 2018 (Hedegaard et al. 2020). For many of these years, increases in opioid overdose mortality rates mapped closely with increasing rates of opioid prescriptions and nonmedical

opioid use (Compton et al. 2015; Han et al. 2015). Studies estimate that as many as 25% of individuals who die from opioid overdose – be it from prescription or illicit sources – had multiple simultaneous opioid prescriptions from different providers (Martyres et al. 2004; Peirce et al. 2012; Yang et al. 2015). Given that opioid abusers tend to be geographically clustered (McDonald & Carlson 2014; Nordmann et al.), and tend to obtain opioids from relatively narrow sets of lax or complicit opioid providers (Barnett et al. 2017; McDonald & Carlson 2013; Stein et al. 2017), it is believed that patterns of nonmedical opioid use correspond closely to patterns of deliberate prescription drug-seeking through patient social networks (Wilsey et al. 2010). In other words, it is likely that users of nonmedical opioids leverage their social networks to identify and seek prescriptions from a small subset of opioid prescribers.

Although visiting health care providers to obtain opioids for nonmedical use is a critical predictor of opioid overdose, identifying this behavior among patients is a difficult task that has generated a sizable body of research (Cepeda et al. 2012; Pradel et al. 2009; Sansone & Sansone 2012; Wilsey et al. 2010). One common measurement tracks whether a patient had a multiple provider episode (MPE), defined by obtaining opioid prescriptions from multiple doctors simultaneously (commonly four) over a short period of time (commonly 90 days) (Cepeda et al. 2012; Pradel et al. 2009). However, classifying prescription drug-seeking using MPE results in estimates of prescription drug-seeking with high specificity but low sensitivity; very few patients obtain opioid prescriptions from multiple prescribers over a short period of time. Yet, lower thresholds for measuring opioid-seeking through MPEs are equally problematic, as many patients who obtain opioid prescriptions from two or three doctors during a short time period often do so for legitimate (and often urgent) medical conditions.

A second common measurement for prescription opioid-seeking is based on the potency, or the total morphine milligram equivalent (MME), of all opioids prescribed to a particular patient during a particular interval of time (Morris et al. 2014). However, MME does not always track closely with the true abusability of a patient's prescriptions. Many opioids have high MME but have relatively low abuse rates due to the distinct pharmacological properties of those drugs. For example, buprenorphine has high MME, but its slow absorption rate and its tendency to block certain opioid receptors in the brain substantially reduce its popularity among nonmedical opioid seekers. In fact, these properties make buprenorphine an effective medication for opioid use disorder (Thomas et al. 2014). In addition, many individuals that receive a large total MME of opioid prescriptions obtain them from only a single doctor, and many prescribers of high MME drugs are specialists (e.g., oncologists) that are unlikely to be identified and sought out by people that seek opioids for nonmedical use.

Because prescription drug-seeking is a relational behavior, several studies have proposed using social network analysis to identify it (Ong et al. 2016; Perry et al. 2019; Stein et al. 2017; Takahashi et al. 2016). Nonmedical prescription drug seekers typically seek out providers who are highly willing to prescribe opioids, either because the providers are unaware of the risks of opioid addiction, have poor training in identifying appropriate analgesic use, are easily duped by fraudulent patients, or because the providers are directly engaged in illegal drug diversion (Perry et al. 2015; Wesson & Smith 1990). For this reason, patients seeking prescriptions for nonmedical are believed to visit these same sets of prescribers and to recommend such prescribers to each other (McDonald & Carlson 2014; Perry et al. 2019; Takahashi et al. 2016). In aggregate, shared preferences for prescribers and the tendency to find prescribers through patient referral networks should result in clustering of drug-seeking patients around the same sets

of generous opioid prescribers. This also results in patients seeking opioids for nonmedical use holding prominent structural positions within patient-provider opioid prescription networks (Perry et al. 2015).

Recently, two studies found that network centrality in a patient-provider opioid prescription network is strongly associated with other common measures of nonmedical prescription opioid use and that centrality is significantly more predictive of future opioid overdoses than are traditional measures (Perry et al. 2019, 2020). Though these studies are an important first step, and an improvement over individual-level proxy indicators of patient behavior, they only measured network centrality with PageRank on a one mode projection of patient-provider networks. Consequently, they may produce less accurate estimates than bipartite centrality measures.

In the following sections, we describe a series of alternative centrality measures for estimating prescription drug-seeking behavior. We then predict the degree to which the algorithms reflect true prescription opioid-seeking behavior. Differences in how the various bipartite ranking algorithms assign centrality scores to nodes should also underscore their pros and cons across network settings, and help us identify which algorithms are likely to be most generalizable across contexts. Throughout our comparison of these methods, we consider the degree to which each centrality measure aligns with our four expectations of nonmedical prescription-seeking behavior: 1) Seeking opioids through multiple prescribers simultaneously; 2) Visiting generous providers of opioid analgesics; 3) Clustering around the same set of generous opioid prescribers; 4) Visiting providers that prescribe higher quantities of potent opioids.

3. Centrality in Bipartite Social Networks

3.1. PageRank Centrality on One Mode Network Projections

PageRank is a specialized variant of eigenvector centrality that measures the extent to which a node is connected to other prominent nodes in a social network (Friedkin & Johnsen 2014; Page et al. 1999). PageRank was originally created as the backbone of the Google internet search engine and was developed based on a conceptual model of millions of individuals surfing through webpage links on the internet. Specifically, PageRank identifies how often different webpages would be reached if web-surfers were to click randomly across website links. Essentially, PageRank estimates the probability distribution for all websites to be found through thousands of random walks across the website network (Melamed 2014). Nodes that are reached more frequently through these random walks are assigned a higher PageRank.

The algorithm for PageRank is formalized as follows:

$$R(i) = \delta \sum_{j \in M(i)} \frac{R(j)}{k_j^{out}} + (1 - \delta) R^{0(i)}, \quad (1)$$

$R(i)$ represents the PageRank of node i, δ indicates a damping factor assigned to the random walk (typically set at 0.85), $M(i)$ represents the set of nodes that point to node i, k_j^{out} indicates the out-degree of node j, and $R^{0(i)}$ is typically set to $\frac{1}{N}$ where N represents the population of the network. The algorithm first divides node ranks by the degree of each node $\frac{R(j)}{k_j^{out}}$ to normalize the calculation; without this specification, PageRanks would diverge across every iteration of the algorithm. In matrix notation, if **R** represents a vector of PageRank values for every node and *S* represents the normalized transition matrix, PageRank can be expressed as:

$$\mathbf{R} = \delta S \mathbf{R} + (1 - \delta) \mathbf{R}^0 \quad (2)$$

Although PageRank and other variants of eigenvector centrality were designed for use on unipartite graphs, these algorithms are frequently applied to studies of bipartite social networks. To estimate PageRank centrality on a bipartite social network, the original network must first be converted to a unipartite social network. The most common methods for converting bipartite networks to unipartite networks are to (1) take the cross product of the original network such that the resulting network represents transitive ties between nodes on a single mode of the original network (Latapy et al. 2008; Newman et al. 2001; Uzzi & Spiro 2005; Wasserman et al. 1994), (2) to treat the network as if ties were possible within modes by adding empty rows for each node that was only represented in the columns of the original network and vice-versa (Bonacich 1991; Borgatti & Everett 1997), or to (3) dual project the social network by estimating ranks on each mode of the network separately before combining the network back into its original form (Everett 2016; Everett & Borgatti 2013; Grassi et al. 2019). These conversion methods can interact with some centrality algorithms differently than others, but for PageRank and other variations of eigenvector centrality, each method returns nearly identical node ranks (Bonacich 1991). At the same time, such conversion methods always result in some degree of network distortion that can bias centrality estimates or even give rise to wholly spurious network structures (Newman et al. 2001; Guillaume & Latapy 2004). In particular, estimating PageRank on a bipartite social network requires one to assume that ties were possible between nodes within the same mode of the network.

Within a patient-provider network, we expect PageRank to measure a patient's network centrality and likely prescription drug-seeking behavior to a large extent, albeit somewhat imprecisely. PageRank is likely to assign high centrality estimates to patients connected to frequent opioid prescribers and to patients who see the same sets of doctors as other likely drug-

seeking patients. However, PageRank on the one mode projection of a patient-provider network will treat all transitive ties as having the same structural importance. This means that a patient with only one provider tie could have the same outdegree in the projected network as a patient who has ties to multiple weakly-connected doctors. Since PageRank is highly influenced by a node's outdegree, this feature of a one mode network projection is likely to significantly bias how well PageRank tracks onto true prescription drug-seeking behavior. We expect some patients with high PageRank estimates to have connections to only one or two well-connected providers, and we expect some patients with low PageRank estimates to have connections to larger numbers of poorly-connected providers. Both of these outcomes are exactly the opposite of how we would typically want to define and operationalize patients who visit multiple providers to obtain opioids for nonmedical use. In addition, because PageRank on the one mode network projection will not respect the bipartite topology of the patient-provider network, we expect PageRank to distort estimates of patient behavior in other subtle but not necessarily predictable ways.

It is worth noting that there are many edge-weighting methods for overcoming the tendency for PageRank on one-mode network projections to overestimate the centrality of low-degree nodes in the original bipartite network. For example, inverse hyperbolic weights reduce the strength of any valued tie on the inverse product of each node's degree (Newman 2001). Inverse hyperbolic weights for a tie between node *i* with an outdegree of 4 and node *j* with an outdegree of 3 would be equal to $\frac{1}{3} * \frac{1}{4} = \frac{1}{12}$. The intuition behind inverse hyperbolic weights is that nodes with many ties might have a reduced tie strength when they have more connections. In a student-teacher network, for example, it is likely the amount of time teachers devote to each of their students is inverse to the number of students in each teacher's class. The problem with

inverse hyperbolic weights and all weighting methods, however, is that they, too, come with their own assumptions and distortions of the original social network. In patient-provider networks, inverse hyperbolic weights assume that well-connected providers are more weakly tied to their patients than poorly-connected providers are tied to their patients. This assumption is likely invalid. It is unclear how inverse hyperbolic weights would interact with PageRank, but one possibility is that it could result in lower centrality scores for patients connected to well-connected providers than to patients connected to the same number of poorly connected providers – an exact inverse of what we expect from true drug-seeking behavior. Due to our focus on comparing bipartite ranking algorithms, we do not explore the various edge weighting strategies of one mode projections in further detail. Our analysis only examines PageRank when applied to the direct one-mode network projection of the patient-provider network.

3.2. Bipartite Centrality Algorithms

Bipartite centrality algorithms are a direct extension of the PageRank algorithm and follow the intuition that a vertex should have higher centrality if it is connected to other vertices with high centrality. Like PageRank, bipartite centrality algorithms update centrality estimates for nodes iteratively based both on each node's connection to each other and based on the most recent rank estimates of each node's alters. However, bipartite algorithms differ from PageRank in their ability to estimate centrality directly on a bipartite network simultaneously for both modes of the network. Bipartite centrality algorithms respect the impossibility of ties between nodes within the same mode of the social network, and the necessity for centrality to only compare nodes within the same mode of a network. In every step of the Markov process, bipartite centrality algorithms update centrality estimates for every node in the social network based on – their connections to other nodes in the social network, their centrality estimates in the

previous stage of the algorithm, and by their connections' centrality estimates in each stage of the algorithm. Put together, bipartite centrality algorithms cause centrality estimates to propagate back and forth across each mode of the network throughout each iteration of the algorithm.

The bipartite centrality algorithms tested in this study are HITS, CoHITS, BGRM, and BiRank (Deng et al. 2009; He et al. 2017; Kleinberg 1999; Rui et al. 2007). These algorithms were developed separately and for distinct purposes, but many of their underlying similarities have been identified and systematized in recent work by He et al. (2017). The algorithm underlying all bipartite centrality measures is formalized as follows:

$$p(j) = \alpha \sum_{i=1}^{D} \frac{w_{ij}}{S^*_{p(i)}} d(i) + (1 - \alpha) p^0(j),$$

$$d(i) = \beta \sum_{j=1}^{P} \frac{w_{ij}}{S^*_{d(j)}} p(j) + (1 - \beta) d^0(i) \quad (3)$$

Here $p(j)$ and $d(i)$ represent the bipartite centrality of patient j and provider i separately, α and β represent the damping factors for the random walk (typically set to 0.85), D and P represent the number of providers and patients in the network, w_{ij} indicates the element of the adjacency matrix W^{D*P} at coordinate (i, j). $S^*_{p(i)}$ and $S^*_{d(j)}$ represent normalizers that differ across each bipartite centrality algorithm and are outlined further below. In matrix notation, if **p** and **d** represent a vector of centrality estimates of every node in each mode of the network and S_p and S_d represent transition matrices of the network across iterations, the bipartite centrality algorithm can be expressed as:

$$\mathbf{p} = \alpha S_p \mathbf{d} + (1 - \alpha) \mathbf{p}^0,$$

$$\mathbf{d} = \beta S_d \mathbf{p} + (1 - \beta) \mathbf{d}^0 \quad (4)$$

As alluded to above, the variations of the bipartite centrality algorithm differ primarily in how they normalize the social network prior to iteration. Moreover, we expect some variations of the bipartite centrality algorithm to reflect patient prescription-seeking behavior better than others.

Kleinberg's (1999) HITS (Hyperlink-Induced Topic Search) algorithm is the oldest of the four bipartite centrality algorithms tested here, and is the least similar to the other bipartite algorithms. HITS was originally designed as an alternative to PageRank for estimating node prominence in one mode networks by iteratively ranking nodes according to their role as an "authority" in the network and their role as a "hub" in the network. Nodes with high authority scores are defined by having a high indegree from high ranking hubs; and nodes with high hub scores are defined by having a high outdegree to nodes with high authority. The version of HITS tested here was slightly expanded by He et al. (2017) to interface with bipartite social networks, but is otherwise identical to the algorithm proposed by Kleinberg (1999). HITS differs from every other bipartite centrality algorithm in that it does not normalize the input adjacency matrix in any way prior to initializing its Markov process. Indeed, to ensure convergence, HITS normalizes centrality estimates after every iteration of the Markov process by dividing each node's rank by the sum of all ranks on that mode. In one sense, HITS requires the least network distortion of any bipartite centrality algorithm since it does not transform its input in any way prior to initializing its Markov process. However, HITS is known for its tendency to produce unintuitive centrality estimates, even in comparison to PageRank (Devi et al. 2014). There are a variety of network features that can lead to this problem, but HITS' most central weakness is its sensitivity to outliers. Nodes with high degrees can have an extreme and often unexpected

influence on how HITS ranks all other nodes in the network. Based on prior evidence alone, we expect that:

Hypothesis 1: HITS performs worse than PageRank in reflecting patterns of prescription drug-seeking behavior.

CoHITS (Deng et al. 2009) is named for its ability to better-incorporate content information into its algorithm than HITS (the "Co" in CoHITS). In other words, CoHITS estimates reflect the probability distribution of finding nodes through a random walk through the network, whereas HITS estimates have a more abstract interpretation. Like all tested bipartite centrality algorithms other than HITS, CoHITS normalizes the inputted network prior to the first step of the generalized bipartite centrality algorithm. CoHITS, in particular, normalizes edges by the sums of their columns (i.e., their indegree), thereby lessening the contribution of connections to high-degree vertices to each node's centrality score. A possible weakness of CoHITS is that it only normalizes the network on one vertex; nodes with high outdegrees might still exert an undue influence on CoHITS centrality estimates (He et al. 2017). However, we are less certain about this potential weakness, at least with regards to estimating centrality in a patient-provider network. While CoHITS does not directly normalize the network by each node's outdegree, the fact that bipartite centrality algorithms propagate centrality estimates back and forth across each mode of the network may provide an indirect means of outdegree normalization. Moreover, our patient-provider network does not have many extreme outliers, and most outliers only exist on one mode of the network (providers). Put together, these factors lead us to expect that:

Hypothesis 2: CoHITS performs better than PageRank in reflecting patterns of prescription drug-seeking behavior.

BGRM (standing for bipartite graph reinforcement model) was developed for automating web image annotation (Rui et al. 2007), and differs from CoHITS centrally in its use of a symmetrical weighting scheme, whereby all edges are normalized by both vertex ends. In this sense, BGRM normalizes the input matrix much akin to inverse hyperbolic weighting by dividing each edge of the input adjacency matrices by both the degree of the source node and the degree of the target node. As noted above, we are wary of inverse hyperbolic weighting in the patient-provider network due to the extreme extent to which it normalizes the network. We expect BGRM's normalization scheme to be too strong for the patient-provider network. In particular, we believe BGRM will assign overly low centrality estimates to high degree patients that are connected to high degree providers and overly high centrality estimates to low degree patients that are connected to low degree providers. For this reason, we expect that:

Hypothesis 3: BGRM performs worse than PageRank in reflecting patterns of prescription drug-seeking behavior.

BiRank is the newest of the bipartite centrality algorithms and was developed to improve upon the theoretical advantages of BGRM's symmetric normalization scheme (He et al. 2017). BiRank divides each edge by the square root degree of the source node and the square root degree of the target node prior to iteration. This means that BiRank exerts a significantly lower degree of normalization on the inputted network than BGRM, and in theory, should be less prone to outliers on either mode of the network than CoHITS. However, we are uncertain to what extent BiRank's symmetric normalization scheme improves upon CoHITS for estimating patient behavior. Since there are few extreme outliers on the patient mode of the network, BiRank might not yield any noticeable advantages over CoHITS. We therefore only expect that:

Hypothesis 4: BiRank performs better than PageRank in reflecting patterns of prescription drug-seeking behavior.

A summary of the differences among the bipartite centrality algorithms' normalization schemes are formalized in Table 1.

Table 1
Summary of Bipartite Centrality Algorithms

Algorithm	Transition Matrix of Top Mode (S_p)	Transition Matrix of Bottom Mode (S_d)
HITS	W^T	W
CoHITS	$W^T K_d^{-1}$	$W K_p^{-1}$
BGRM	$K_p^{-1} W^T K_d^{-1}$	$K_d^{-1} W K_p^{-1}$
BiRank	$K_p^{-1/2} W^T K_d^{-1/2}$	$K_d^{-1/2} W K_p^{-1/2}$

K_d and K_p represent diagonal matrices with the generalized degrees (sum of edge weights) on the diagonal of each mode of the network. Specifically, $(K_d)_{ii} = \sum_j w_{ij}$ and $(K_p)_{jj} = \sum_i w_{ji}$.

3.3. Edge Weights

In most studies of social networks, centrality scores are not meant to reflect an individual's literal node prominence, but, rather, to reflect some latent position in their social hierarchy. In our current study, we are using centrality to measure patients' degree of drug-seeking behavior. However, there are still some ways in which centrality is unlikely to closely represent this behavior. Although bipartite centrality algorithms reflect the tendency for patients to seek multiple providers simultaneously, to seek generous opioid prescribers, and to seek providers that are sought by other patients seeking opioid prescriptions, bipartite centrality is unlikely to reflect patient preferences for providers that administer large quantities of potent opioids.

To capture patient preferences for providers that administer large quantities of potent opioids, we propose an innovative adaptation of prescription drug-seeking measures. Specifically, we weight provider-patient ties by a potentially relevant edge trait – the total morphine milligram equivalent (MME) of opioids prescribed to the patient. MME is a common measurement for estimating the potency of opioid drugs that is based on the equivalence of an opioid prescription with 10mgs of orally taken morphine. MME typically refers to the potency of a single dosage of an opioid, but some scholars examine the total MME of all pills prescribed to a particular patient to estimate the potency and quantity of opioids prescribed, and occasionally, to track potential fraudulent behavior and nonmedical use (Morris et al. 2014). By weighting each edge of the patient-provider network by the total MME of prescriptions administered by a provider to a particular patient, we improve our measurement of centrality by incorporating patient preferences for providers that administer large quantities of highly potent opioids. This strategy should be particularly effective for reducing the centrality estimates of patients who are connected to providers who only administer short term analgesic treatments, such as surgeons and emergency care physicians.

Although MME weights will likely lead to substantial changes in our centrality estimates, we expect the resulting centrality measures to better reflect true drug-seeking behavior and better predict subsequent opioid overdose than centrality measures without MME-weighted edges. Likewise, even though the degree distribution of the social network is likely to change with MME-weighted edges and become more skewed, we still expect the accuracy of the various bipartite centrality algorithms to follow a similar hierarchy to their unweighted counterparts. That is, we expect BiRank and CoHITS on the MME-weighted networks to offer the best estimates of patient drug-seeking behavior. Our tests of bipartite centrality algorithms with

MME-weighted edges should also serve as an additional test of these algorithms' generalizability across different network structures. Algorithms that perform well on the unweighted network but poorly on the MME-weighted network likely have limited generalizability across use cases. Put together, we hypothesize that:

> *Hypothesis 5: MME-weighted edges will improve the accuracy of BiRank and CoHITS in reflecting prescription drug-seeking behavior.*

4. Methods

4.1. Data

Data are drawn from Optum's de-identified Clinformatics® Data Mart Database, a national, commercial, and Medicare Advantage claims database that contains dated information about patient medical histories, office appointments, and prescriptions during the years 2007 to 2018. The sample does not include medical claims across all providers, but there are many regions where the medical claims database covers the plurality of patient-provider visits. We lose coverage of patients who switch medical plans over time; however, most patients have multiple years of data in the sample. Due to our theoretical interest in opioid prescription drug-seeking, we limit our sample to patients who have ever received an opioid prescription, identified via their national drug codes (NDCs). Patient-quarters (three-month intervals) comprise our unit of analysis.

To ensure an optimal context for comparing different measures of patient behavior, we focus our analysis to the years 2009 through 2012. We choose 2009 through 2012 because these years had very high rates of prescription opioid abuse (Compton et al. 2016, p. 201; Han et al. 2015). By 2012, many states implemented prescription drug monitoring programs (PDMPs) that

curtailed rates of fraud and abuse, and contributed to a rise in consumption of more easily accessible illicit opioid substances that are not prescribed by health professionals (Ali et al. 2017; Lilly 2019). In other words, focusing on the years 2009 to 2012 reduces the degree to which use of illicit opioids (e.g., heroin, fentanyl) is likely to create noise in our predictions of opioid overdose via prescription drug-seeking.

We also focus our analytic sample on a narrow region within in the U.S. Appalachian Mountain range. Appalachia is a well-known hotspot of opioid overdoses in the U.S. Moreover, narrowing analysis to a small geographic region reduces the complexity of our methodological approach. Specifically, preference for physical proximity plays a large role in which providers a patient chooses to seek opioid prescriptions from. Estimating centrality across a social network that spans a broad geographic region would likely induce network endogeneity that biases the degree to which centrality reflects true drug-seeking behavior. Although there are a variety of methods for accounting for network endogeneity (Bahmani et al. 2010; Boucher & Fortin 2016; Fujimoto et al. 2011; Mouw & Entwisle 2006), narrowing our analysis to a small geographic region is the most parsimonious option. Since we are primarily concerned with comparing different measures of centrality, we endeavor to keep our analytic approach as straightforward as possible. The geographic region of our analytic sample and corresponding rates of opioid overdose within Appalachia are illustrated in Fig. 1. The boundary of the analytic region is composed of county lines across five states, including West Virginia, Virginia, North Carolina, Kentucky, and Tennessee. In total, our subsample within this region and time period includes 245,133 unique patients and 1,920,554 patient-quarters.

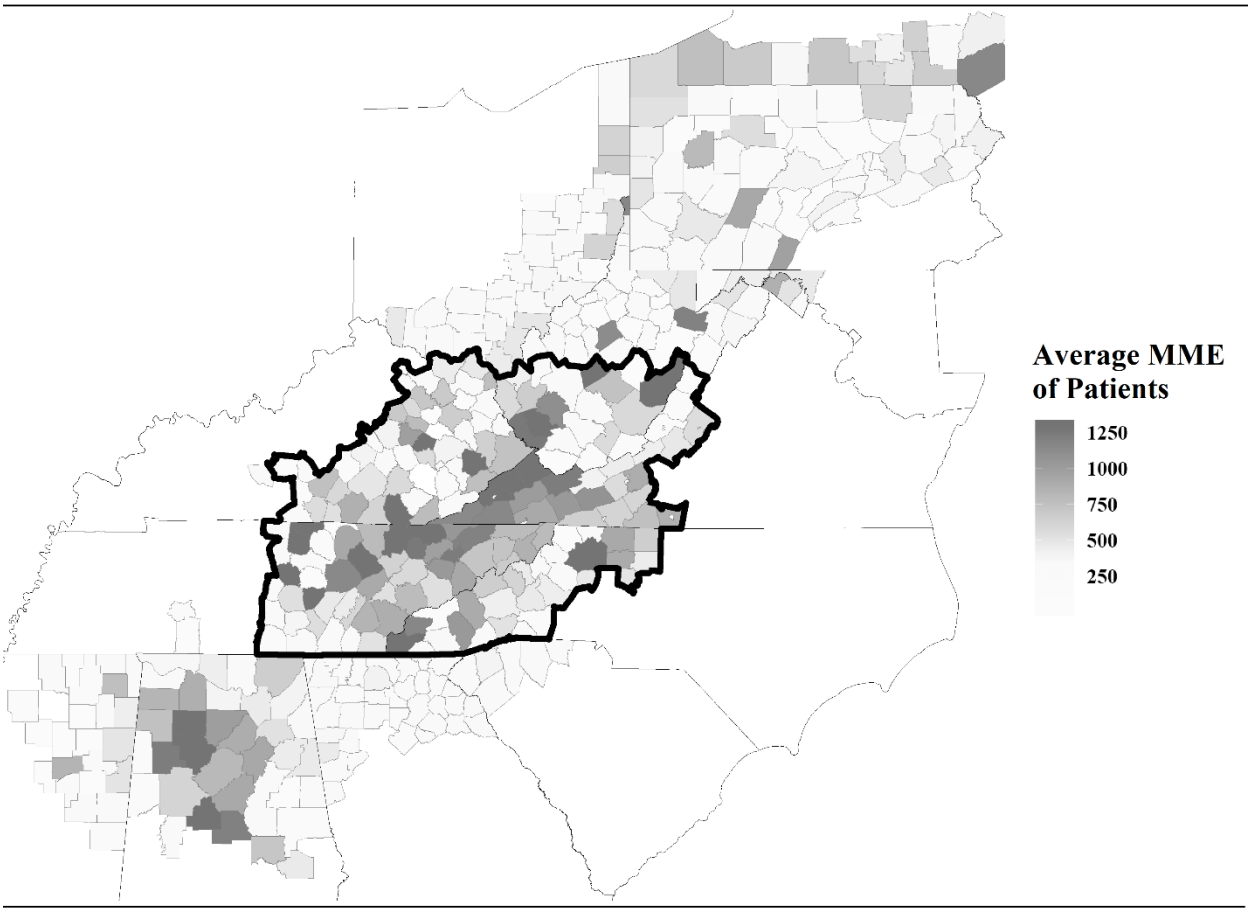


Fig. 1. Analytic Sample within Appalachia. Thick borders indicate sample boundaries; medium borders indicate state boundaries; thin borders indicate county boundaries. Average MME of patients shown for all counties in Appalachia and estimated from the first quarter of 2011.

4.2. Network formation

We use separate intervals from the opioid prescription dataset to construct a longitudinal bipartite social network of patient-provider ties. Ties are defined by whether a patient received at least one opioid prescription from a provider during a given interval. Repeated ties are not treated as valued in the unweighted bipartite network, but we do give weight to repeated transitive ties in the one mode network projection with PageRank centrality, and we indirectly give weight to repeated ties in our secondary analyses of the MME-weighted patient-provider network. Although our unit of analysis is patient-quarters, we construct the longitudinal bipartite

social network with three-quarter (nine month) rolling windows of patient-provider prescriptions. We made this choice to better reflect the duration of true drug-seeking behavior. With one quarter intervals, we noticed that some patients exhibited periods of limited behavior directly preceded and followed by extreme periods of prescription drug-seeking. These short gaps do not have any clear negative association (or lagged negative association) with reduced rates of opioid overdose. We believe it unlikely that such patients truly stopped seeking drugs during these short intervals. Rather, we believe these patients either had short periods of difficulty in successfully attaining opioid prescriptions or short periods of reprieve in visiting prescribers due to their recent successes in attaining drugs either legally or illegally. However, to ensure that our analyses are not biased by our choice to construct the patient-provider network with three-quarter rolling windows of prescription data, we re-estimated all models with networks comprised of only a single quarter of prescription data, and found that these yielded substantively identical results to those constructed using three quarters.

4.3. Statistical models

We primarily compare different centrality estimates of prescription drug-seeking behavior by examining their ability to predict subsequent opioid overdose. We estimate the degree to which each independent and control variable is associated with a patient's conditional probability (hazard) of opioid overdose over time with Cox proportional hazard models (Cox & Oakes 1984). Cox proportional hazard models are a type of event history model that are often used for studying the association between uncommon events with multiple predictors (Allison 1984). One key advantage of Cox models over traditional regression methods is their ability to prevent censorship bias. Many patients were dropped from the sample for reasons unrelated to opioid overdose (often because they switched employers and insurance providers), but others

died from opioid overdose and were censored due to their death. If we used logistic regression rather than Cox regression, censorship would cause bias in parameter estimates because the missing information is related to the dependent variable. In contrast, Cox models prevent censorship bias by removing censored observations from the risk set; censored individuals only contribute to hazard estimates during the years in which they were present. A second key advantage of Cox proportional hazard models is that they are semi-parametric. Traditional regression methods assume that data points lie across a particular distribution of event-times, be it normal, logistic, or Poisson, but Cox models make no such assumptions. This is critical for our analysis because we have no *a priori* hypothesis about the distribution of opioid overdose event times. Overdose events are rare and are unlikely to follow any typical probability distribution.

To compare the extent to which different measurements of centrality are representative of prescription drug-seeking behavior and predictive of opioid overdose, we estimate separate Cox proportional hazard models for each measurement of centrality. We manipulate each model specification only by the type of centrality measure included to identify patient behavior; we do not include all centrality measures at once because the measures are highly collinear. We compare the degree to which each model fits the data based on their resulting Akaike Information Criterion (AIC) (Sakamoto et al. 1986). AIC measures the maximum likelihood of each model while penalizing for each additional estimated parameter, such that models that are overfit will have worse AIC than models with fewer parameters. Because AIC measures a model's maximum likelihood, a model with a substantially lower AIC than another model can be statistically shown to fit the data better than the other. For this reason, we expect that AIC is likely to help indicate which centrality measure best reflects true prescription drug-seeking behavior.

4.4. Variables

The dependent variable across models is opioid overdose, defined by whether a patient was marked with a corresponding International Classification of Diseases 9th edition code (ICD-9) (World Health Organization 1978). Given that opioid overdose events are rare and the substances underlying opioid overdoses are often mixed and ambiguous, we do not attempt to distinguish patients who had overdosed on illicit or unspecified narcotic substances from those who had overdosed on prescribed opioids. To improve our ability to estimate the causal effect of patient behavior on opioid overdose and account for the tendency of prescription drug-seeking to precede opioid overdose, we also lag this dependent variable by one quarter. In total, our sample includes 390 overdose events.

The key independent variable across models is patient centrality. Centrality is estimated with PageRank on the one mode network projection, HITS, CoHITS, BGRM, and BiRank via the birankr statistical package in R (Aronson & Yang 2020). We normalize all centrality measures across network components to keep centrality estimates at a consistent scale, and we take the natural log transformation of each centrality measure to account for the natural right-skewed distribution of centrality estimates. However, to ensure that the choice to take the natural log transformation of each centrality measure did not bias our estimates, we also estimated all models with centrality measures that were left untransformed and with centrality measures that were dichotomized above and below the 99th percentile (available upon request). Across all transformations of centrality, each model yielded substantively similar results.

We include a variety of independent variables to control for any spurious relationship between centrality and opioid overdose. We include age (in years) and sex (female) to reflect the slightly increased propensity for opioid overdose by people of middle age and male sex. We

examine two individual-level network parameters that are likely associated with patient drug-seeking behavior, including patient degree (number of unique providers) and transitive ties (number of connections to other patients through one's providers). These network parameters are likely to be associated with patient behavior, but neither of these parameters closely reflect a patient's position within the entire structure of the patient-provider network. More specifically, neither network parameter reflects the tendency for patients to seek generous providers or to cluster around the same sets of generous providers. Excluding these individual-level network parameters generated more statistically significant parameter estimates for network centrality; however, we include these parameters as an indirect test for whether network centrality even provides any distinct contribution to predicting opioid overdose beyond traditional measures of drug-seeking behavior based on individual-level network characteristics. Finally, we control for patients who are likely to receive more opioid prescriptions due to having a particular disease rather than due to nonmedical opioid use. These controls include whether a patient had hepatitis C (HEPC), human immunodeficiency virus (HIV), cancer, a psychiatric disorder, was receiving medically assisted treatments (MAT) for opioid addiction, or whether the patient was in palliative care.

5. Results

To orient readers to the analytic sample, we first illustrate the largest and second largest connected components of the patient-provider network during the first period of analysis (October 1, 2008 to June 30, 2009) in Fig. 2A and Fig. 2B. The total number of patients sampled during the first period of analysis is 148,182 patients, but during this period, only 41,698 patients received opioids from any doctor and are represented in the network. The largest component of the network comprises 33,024 patients or 79.2% of patients who received any opioids, 8,273

providers, and 49,738 edges, and the second largest component of the network comprises 91 patients or 0.2% of patients who received any opioids, 64 providers, and 164 edges. The high coverage of the largest connected component implies that patient-provider ties in this region of Appalachia represent a relatively coherent and well-connected network. Another notable pattern in Fig. 2 is that the patient-provider network appears relatively diffuse. No nodes appear to have extreme degrees in Fig. 2. Although Fig. 2A contains four somewhat denser clusters of nodes, there are many visible nodes and ties that lie between each cluster.

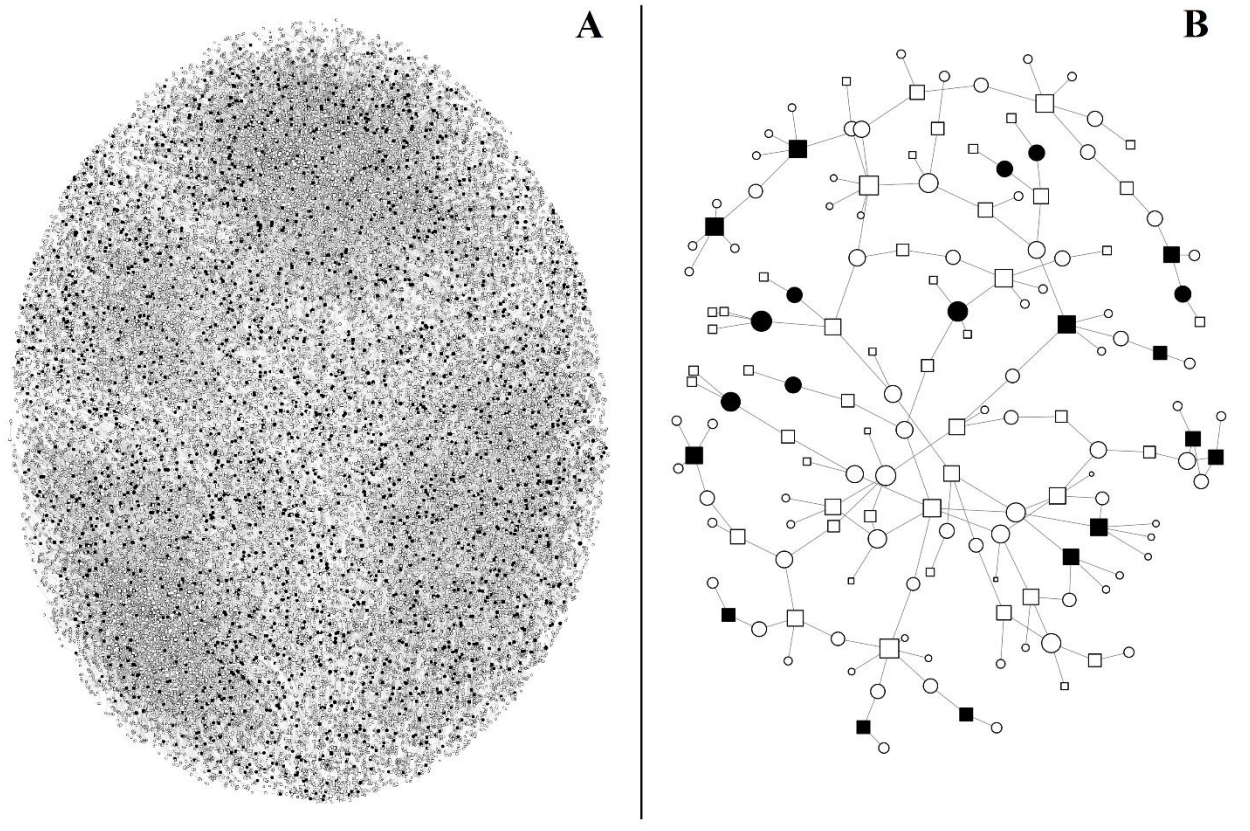


Fig. 2. Sociogram of Largest Two Network Components. **A** indicates largest network component; **B** indicates second largest network component. Network constructed from all opioid prescription ties from October 1, 2008 to June 30, 2009. Nodes are sized according to their BiRank centrality. Circular nodes indicate patients; square nodes indicate providers. Nodes are shaded if their BiRank is two standard deviations above the mean BiRank and if their PageRank is below the mean PageRank.

Nodes on each mode of the network in Fig. 2 are scaled according to their BiRank centrality and shaped according to whether the node represents a patient (circle) or a doctor (square). To emphasize differences between BiRank and PageRank estimates, we shade nodes if their BiRank is two standard deviations above the mean BiRank and if their PageRank is below the mean PageRank. Nodes towards the center of Fig. 2B and within the four clusters of Fig. 2A tend to be grey, implying that PageRank and BiRank tend to assign similar ranks to well-connected nodes. Most shaded nodes appear towards the periphery of each network, implying that BiRank is more likely to assign above mean centrality estimates for seemingly peripheral nodes than PageRank. However, a close examination of Fig. 2B reveals that most patients with high BiRanks and low PageRanks are connected to multiple poorly-connected providers in the global network. These patterns are consistent with our predictions; in comparison to BiRank, PageRank estimates relatively low centrality to high degree patients that are connected to low degree providers.

Differences between each centrality measure are further illustrated in the correlation matrix of our full sample presented in Fig. 3. Most centrality algorithms are moderately and positively correlated; however, HITS has a fairly low correlation with all other centrality methods and BGRM has a negative correlation with PageRank. BGRM's negative correlation with PageRank is also reflected in its negative correlation with transitive ties, the latter of which is highly correlated with PageRank. On the other hand, most bipartite centrality measures have weak associations with transitive ties. Further exploration suggests that this weak association may result from a slight tendency towards negative degree assortativity (Pearson's $r = -0.03$) in the provider-patient network; low degree patients tend to be connected to high-degree providers. Degree contributes substantially to bipartite centrality measures, therefore it is understandable

that the negative degree assortatitivity would induce a negative association between bipartite centrality and transitive degree. Another noteworthy correlation is the strong relationship between BiRank and CoHITS (Pearson's r = 0.98). Even prior to log-transforming each variable, the correlation between BiRank and CoHITS is still quite high (Pearson's r = 0.96). This correlation is much stronger than the correlation between BiRank and BGRM, even though BiRank and BGRM both use symmetric normalization whereas CoHITS does not.

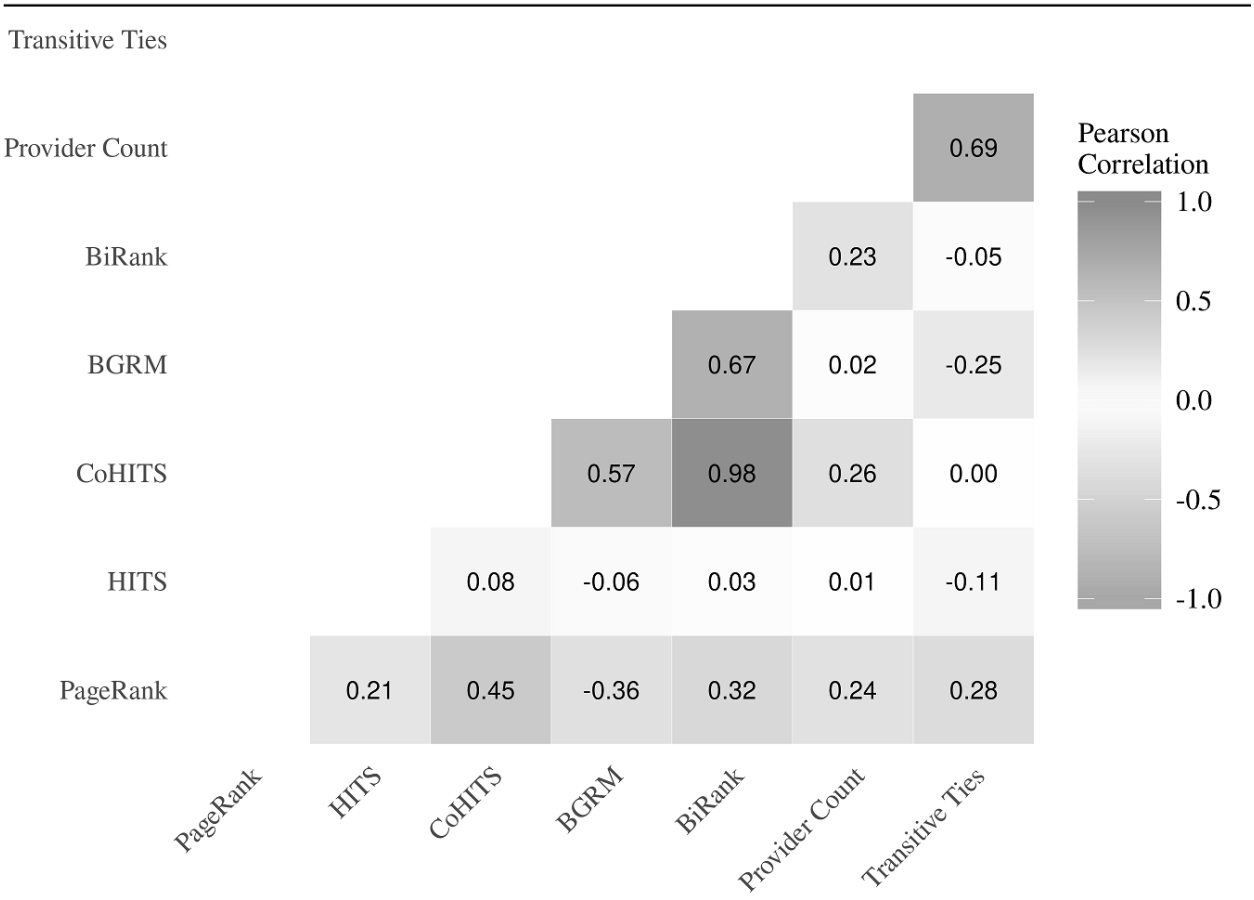


Fig. 3. Correlation Matrix of Key Network Parameters. Pearson correlations estimated from full analytic sample (n = 1,920,554 patient-quarters). All variables are log-transformed.

Fig. 4 illustrates the linear association between each weighted and unweighted centrality measure with opioid overdose. All centrality estimates are only modestly associated with subsequent opioid overdose, but this is expected given the infrequency of overdose events and

the complicated set of conditions that cause such events. The centrality algorithm that produces the greatest direct association with opioid overdose is CoHITS on the MME-weighted network, whereas HITS has the lowest association with opioid overdose. However, it is important to remember that these are raw associations which could be confounded by a variety of factors. The utility of any new benchmark for identifying drug-seeking behavior is largely based on the unique information that the algorithm can provide for predicting opioid overdose beyond that of traditional identifiers. Therefore, these patterns alone should not be interpreted as conclusive evidence that CoHITS on the MME-weighted network is the best predictor of opioid overdose. That said, the extremely limited association of opioid overdose with BGRM and HITS suggests that these algorithms may provide poor direct proxies of prescription drug-seeking behavior.

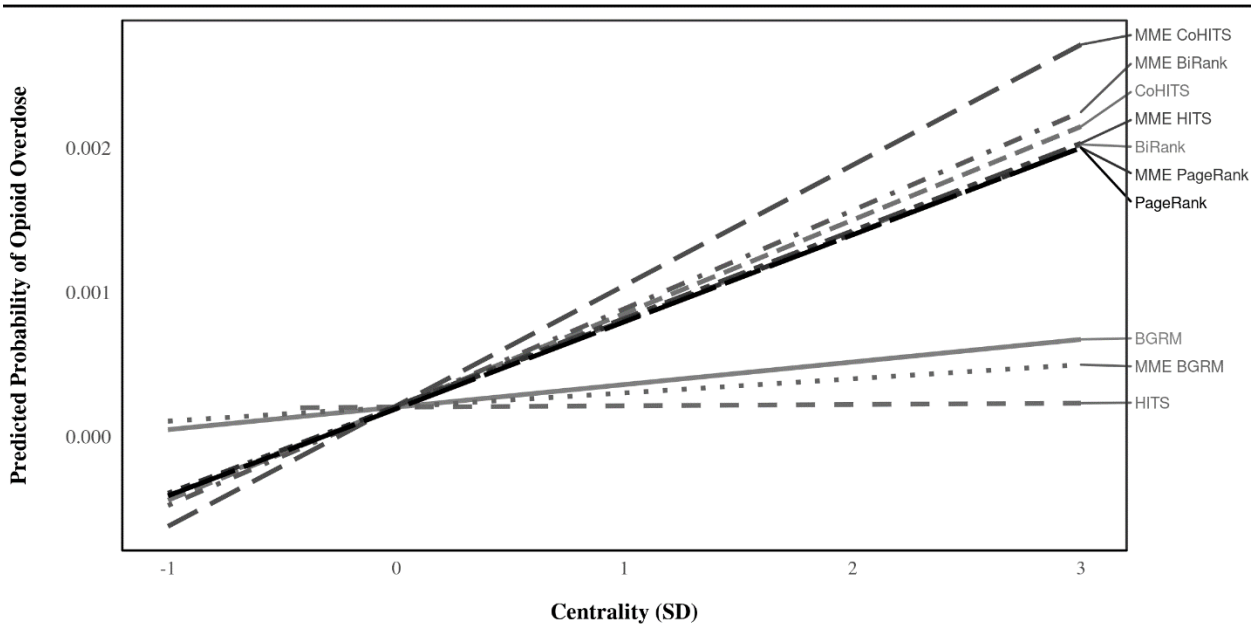


Fig. 4. Linear Association between Opioid Overdose and Network Centrality. Centrality estimates based on patient-provider network of opioid prescriptions. Slopes based on all patient-quarters (1.91m) across analytic period (2009-2012).

Table 2 illustrates five different Cox proportional hazard models that estimate each patients' hazard to overdose on opioids throughout the periods of analysis. Sample statistics

about each measurement are included in the appendix. Across models, each parameter estimate illustrates the hazard ratio associated with a one standard deviation increase in that particular parameter. For example, Table 2 indicates that an individual with a logged BiRank that is one standard deviation higher than another person's logged BiRank has a 1.280 times (or 28%) increased expected hazard of opioid overdose. The five models only differ by their measures of network centrality. Parameter estimates for all centrality scores are positive, but are only statistically significant for CoHITS, BGRM, and BiRank. The other parameter estimates in Table 2 are mostly consistent with expectations. Age and gender are not significantly associated with opioid overdose, whereas degree and transitive ties are both significantly and positively associated with opioid overdose. We observe large increases in expected hazards for opioid overdose among individuals with psychological disorders, on palliative care, or on medication assisted treatments for opioid addiction. The associations of opioid overdose with HEPC, HIV, and cancer are not statistically significant, but the parameter estimate for each variable is positive and large. It is likely that these variables have positive parameter estimates but weak statistical significance due to the limited number of overdose cases in our sample.

Each models' AIC in Table 2 indicates the degree to which the model fits the data. In comparison to the model measuring centrality with PageRank, the BiRank model has an AIC that is (9269 – 9244 =) 25 units lower. This indicates that the model with BiRank centrality is significantly more likely to minimize information loss and improve model fit than the model with PageRank centrality ($p = \exp(-25/2)$; $p < 0.001$). It is tempting to interpret the lower AIC of the model with BiRank as indicating that BiRank provides a better prediction of subsequent opioid overdose than PageRank, but a more direct interpretation is that BiRank provides more relevant unique information to the model than PageRank. Given that BiRank and PageRank have

Table 2
Cox Proportional Hazard Models for Opioid Overdose

	PageRank	HITS	CoHITS	BGRM	BiRank
Centrality	1.065	1.039	1.218 ***	1.338 ***	1.280 ***
Demographics					
Age	0.930	0.924	0.958	0.936	0.963
Female	0.929	0.930	0.930	0.927	0.927
Network Proxies					
Degree (# Providers)	1.844 ***	1.887 ***	1.575 ***	1.668 ***	1.539 ***
Transitive Ties	1.292 ***	1.319 ***	1.419 ***	1.606 ***	1.466 ***
Related Disorders					
HEPC	2.188	2.213 *	1.974	1.946	1.985
HIV	2.678	2.712	2.696	2.510	2.657
Cancer	1.206	1.201	1.168	1.151	1.161
Psych Disorder	7.321 ***	7.306 ***	7.129 ***	7.237 ***	7.106 ***
Palliative Care	4.905 ***	4.995 ***	4.400 ***	4.510 ***	4.376 ***
MAT User	2.524 **	2.439 **	2.569 **	2.407 **	2.538 **
AIC	9269	9269	9248	9242	9244

* = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$. Time-to-event based on one quarter intervals between 2009 and 2012. n = 1.91m patient-quarters. Number of events = 390. Parameters standardized and reported as hazard ratios. a) Number of patient-patient ties through providers. b) Patient is receiving medical assisted therapy.

a similar raw association with opioid overdose, as illustrated in Fig. 4, it is likely that the differences in model fit are attributed to PageRank's higher collinearity with the other model terms (in particular, transitive ties). In other words, PageRank likely better reflects patients' preferences to seek opioids from generous providers of opioid analgesics than BiRank, but PageRank is likely worse at capturing patients' tendency to refer providers to each other and cluster around the same set of generous opioid prescribers.

In line with Hypothesis 1, the model with HITS produces the lowest AIC; however, this model's AIC is not substantially worse than the model with PageRank. Consistent with Hypothesis 2 and Hypothesis 4, models with CoHITS and BiRank have substantially lower AICs than the models with PageRank. Inconsistent with expectations, the model with BGRM has a

smaller AIC than BiRank and CoHITS; however, differences in model fit are very small, the difference in model fit between BGRM and CoHITS barely passes statistical significance at p = 0.05. At the same time, it is important to note that BGRM has a much weaker direct association with opioid overdose than the other network parameters. We ran supplementary models without patient degree and transitive ties, and the model with BGRM was substantially worse than the models with BiRank or CoHITS. In other words, BGRM's weak collinearity with degree and transitivity increases the extent to which BGRM improves model fit relative to BiRank and CoHITS, but reduces its ability to directly proxy patient prescription drug-seeking behavior.

Finally, Fig. 5 illustrates differences between the AICs of our baseline models with the AICs of models that have centrality estimated on the MME-weighted network. The degree distribution of the MME-weighted network is more scale-free than the unweighted network, with a Pearson's moment of skewness at 9.37 in the MME-weighted network and 2.93 in the unweighted network, and a Pearson's kurtosis of 150.44 the MME-weighted network and 16.56 in the unweighted network. In line with Hypothesis 5, models with BiRank and CoHITS centrality measurements on MME-weighted network resulted in lower AICs than their unweighted counterparts. Even though MME-weights appeared to substantially improve HITS' raw association with opioid overdose, the HITS model with MME-weights did not gain any improvement in fit, implying that MME-weights did not increase HITS' ability to provide unique structural information about the network. Also, surprisingly, BGRM performed worse on the MME-weighted network than BGRM performed on the unweighted network. To ensure that the improvement in centrality measures with MME-weighted edges is not simply a result of MME's own direct association with the quantity of drugs consumed, we also estimated models that

included MME as a covariate. However, these results were very similar to those displayed in Fig. 5; therefore, we do not visualize them.

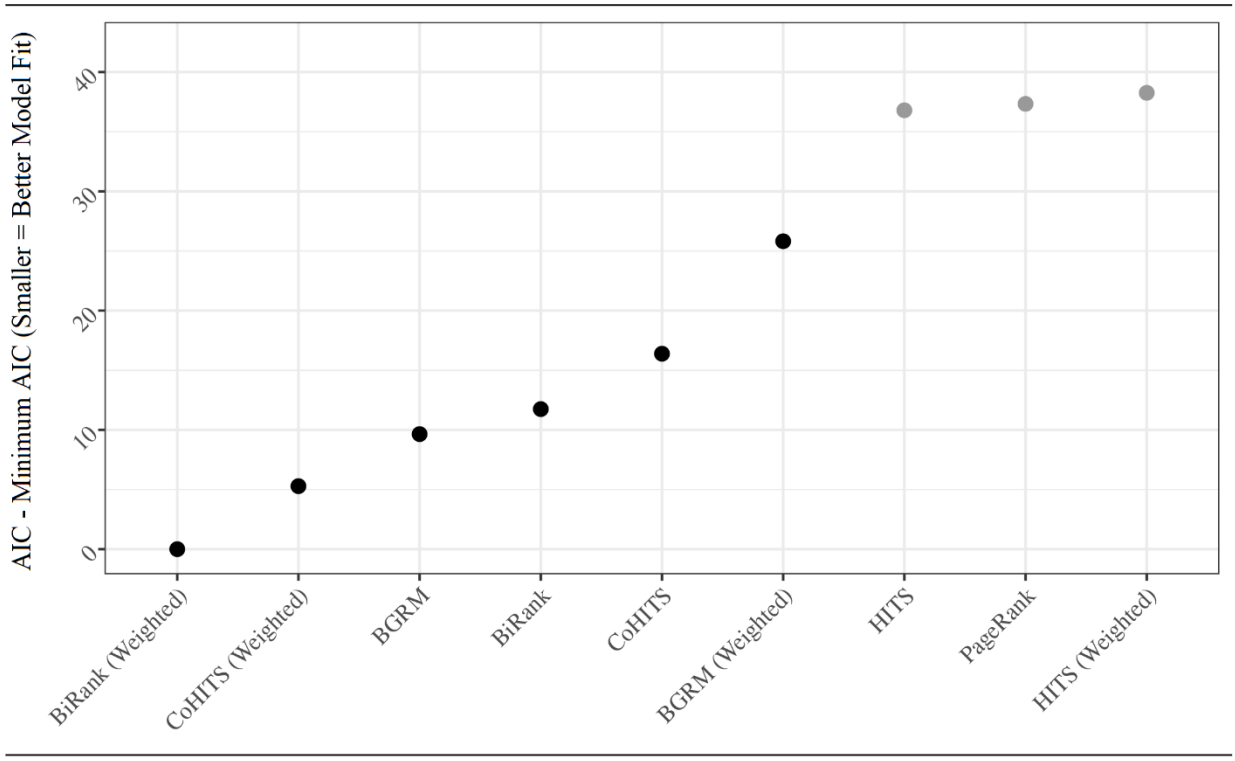


Fig. 5. Model Fit by Rank Parameter and MME-Weighted Edges. Dark points indicate parameter estimates that are statistically significant at $p < 0.01$. All parameter estimates have positive coefficients. Models control for age, gender (female), patient degree (number of providers), transitive ties, HEPC, HIV, cancer, psychological disorders, palliative care, and MAT use.

6. Discussion

In this study, we examined a social network of patient-provider ties defined by opioid prescriptions, and compared how different centrality algorithms reflect prescription drug-seeking behavior and predict subsequent opioid overdose. We conducted this study with three goals: 1) To test the potential advantages of bipartite centrality algorithms in estimating node prominence in bipartite graphs; 2) To provide recommendations as to which bipartite centrality algorithms may be most useful for social network analyses; 3) To suggest a new benchmark for identifying

drug-seeking behavior in medical claims data. Consistent with prior research (Perry et al. 2019; Perry et al. 2020), we expected network centrality to better-reflect the position of patients within the entire social structure of the patient-provider network than individual-level proxies of patient behavior, like degree and transitivity. However, we argued that the extent to which any centrality algorithm represents true prescription drug-seeking behavior is based on the extent to which the centrality algorithm reflects patients' tendency to seek opioids from multiple providers simultaneously, to seek providers who prescribe many opioid analgesics, to recommend and cluster around the same set of generous opioid prescribers, and to seek providers that prescribe higher quantities of potent opioids.

Although PageRank on the one mode network projection is a common method for estimating node prominence in bipartite social networks, we predicted that bipartite centrality algorithms would produce more precise estimates of patient prescription drug-seeking than PageRank due to their closer adherence to our criteria for this behavior. Specifically, we hypothesized that PageRank would overestimate the extent to which drug-seeking is defined by a patient's tendency to seek generous opioid prescribers and underestimate the tendency for patients to seek out multiple providers simultaneously. We also suggested that the ability of bipartite centrality algorithms to account for the bipartite topology of the patient-provider network would increase their ability to detect prescription drug-seeking behavior in other subtle and less observable ways.

Consistent with Hypothesis 2 and Hypothesis 4, we found that PageRank centrality was a worse predictor of subsequent opioid overdose than CoHITS and BiRank. Descriptive analyses supported our expectations that PageRank would underestimate risky behavior among patients with multiple connections to weakly-connected providers. In the context of nonmedical

prescription opioid use, this difference is important; a tendency to receive opioids from multiple prescribers simultaneously is one of the most common measurements of drug-seeking behavior in the literature. While PageRank had a moderate association with opioid overdose, our models suggest that much of PageRank's association with opioid overdose resulted from its ability to capture the tendency for patients to seek opioids from generous opioid prescribers. Including control parameters that could potentially account for PageRank's bias towards assigning overly high ranks to nodes with large numbers of transitive ties still failed to bring PageRank's predictions into parity with those of CoHITS and BiRank. Essentially, PageRank's limited ability to reflect the bipartite topology of the patient-provider network reduced its ability to capture patients' preferences to seek opioids from many providers simultaneously, and this bias likely reverberated into its ability to capture the tendency for these patients to cluster around the same sets of providers.

As expected in Hypothesis 1, HITS performed worse than the other bipartite algorithms in predicting subsequent opioid overdose. As others suggested, HITS' weak performance may be due to its lack of any direct normalization scheme on the bipartite social network (Devi et al. 2014; He et al. 2017). Without normalizing the network, HITS may be too susceptible to outliers. While the direct association between HITS and overdose appeared to improve on the MME-weighted network even though the MME-weighted network had even more degree outliers, HITS' increased association with opioid overdose was entirely mediated by other network controls, suggesting that HITS was still unable to pick up any distinctive structural information about the network. Put together, this suggests that despite its ability to reflect the bipartite topology of the patient-provider network, HITS is still a poor alternative to PageRank in this context.

Contrary to Hypothesis 3, BGRM did not perform worse than PageRank in predicting opioid overdose with the unweighted patient-provider network. In fact, models with BGRM had significantly better fit than models with CoHITS. However, descriptive statistics and supplementary analyses suggested that BGRM's advantage over CoHITS relies on the presence of controls for patient degree and transitive ties. BGRM's direct association with opioid overdose was much smaller than PageRank's direct association with opioid overdose. We consider BGRM's reliance on covariates to be a clear disadvantage in comparison to BiRank and CoHITS. More troubling, BGRM with MME-weighted edges performed worse than BGRM on the network without MME-weights. This implies that BGRM's normalization scheme prevented it from tracking onto patients' preferences for prescribers of potent opioids, and suggests that BGRM does not generalize well to scale-free networks. BGRM also had a negative correlation with transitive ties. Put together, we think these facts imply that BGRM tracks onto patients' tendency to seek multiple providers simultaneously and to cluster around the same sets of opioid providers; however, BGRM does not appear to reflect preferences for generous opioid providers.

As postulated in Hypothesis 5, we found that weighting patient edges by the potency of opioids prescribed by each provider (MME) improves BiRank and CoHITS' ability to predict subsequent opioid overdose. We believe that the MME-weighted centrality measurements improved model predictions because weighting the network by MME better captured patients' preferences for providers that prescribe large quantities of highly-potent opioids. This finding illustrates the potential advantage of weighting patient-provider networks by prescribed MME in studies that seek to identify potential drug-seeking behavior in a patient-provider network, and this finding underlines the potential advantages of weighting bipartite networks by potentially relevant edge traits. Moreover, the improvement of BiRank and CoHITS centrality with MME

weights implies that these algorithms are generalizable to networks with valued ties and to networks with highly skewed degree distributions.

Two of our results may have surprised readers. First, BiRank and CoHITS produced highly correlated centrality estimates despite substantial differences in their normalization schemes. This finding is in opposition to He et al. (2017), who proposed that BiRank's symmetric normalization scheme would provide substantially more accurate estimates of node prominence than CoHITS. We had suggested that these algorithms produced similar estimates because CoHITS indirectly performs a form of symmetric normalization through iterating across both modes of the network, but perhaps these algorithms would produce more disparate estimates on scale-free networks.

Second, we found that the patient-provider network had negative degree assortativity; however, we believe this tendency reflects true patterns of patient-provider relationships. Some types of providers distributed many opioids but were likely to be ill-favored by patients seeking prescription opioids due to the nature of their specialization. For example, emergency care physicians tend to treat many unique patients, but are likely avoided by patients seeking drugs for nonmedical use due to their inaccessibility via appointment. Likewise, surgeons tend to prescribe opioids often, but likely do not attract patients seeking opioids because most surgeons only provide them after invasive medical procedures. We believe removing these specialists would induce a positive degree assortativity in the provider-patient network, but we included these specialists in the network to highlight the advantages of bipartite ranking methods in a real use case. Most scholars do not remove doctors with these specializations from the social network, even though PageRank is likely to over-estimate potential drug-seeking behavior among patients with connections to these high degree specialist providers. Fortunately, bipartite

centrality algorithms can much better account for the limited number of patients engaging in true prescription drug-seeking behavior who obtain opioids from high degree specialists.

Our analysis had a number of limitations. First, our comparisons of centrality algorithms on bipartite social networks was limited to one particular context, but the extent to which any centrality algorithm can accurately identify node prominence likely varies by context. PageRank's bias towards nodes with high degree alters has some clear disadvantages for measuring drug-seeking in patient-provider networks, but these biases might not be so problematic for contexts where the structure of the network has a clearer hierarchy, a more core-periphery structure, or where node outdegree is not necessarily expected to be related to node prominence. For example, PageRank might be advantageous over CoHITS and BiRank for tracking opinion leadership in co-citation networks, since it is arguable that even authors with one or two prominent publications have a great deal of opinion leadership.

Second, although our evidence is based on the extent to which different centrality measures reflect prescription drug-seeking behavior, we do not know who truly engaged in this behavior. Instead, our comparisons were based on the extent to which each proxy measure is associated with subsequent opioid overdose. Although this methodology adheres with prior convention (Perry et al. 2019; Perry et al. 2020), it allows room for our estimates to be biased by network features that are related to opioid overdose but are not related to patient drug-seeking behavior.

Third, it is unclear whether our estimates of prescription drug-seeking behavior are biased by missing information. Our medical claims database does not contain all patients within our region of analysis. However, centrality algorithms that measure node prominence tend to be robust to missing information (Smith & Moody 2013; Wang et al. 2012).

Finally, we only presented findings from a context where prescription drug-seeking behavior is likely to be abundant and where noise from illicit opioid distribution networks is likely to be limited. From the evidence we provided in this manuscript, it is unclear whether our measures of BiRank or CoHITS centrality would accurately estimate patient behavior in more recent medical claims data or in other regions of the country. Addressing this question was deemed beyond the scope of our analysis; nonetheless, some of our current exploratory work suggests that our findings are consistent in contemporary national data.

7. Conclusion

This article compared various measures of node prominence in patient-provider networks to identify potential prescription drug-seeking behavior for nonmedical use or diversion. Our evidence suggests that BiRank and CoHITS may be excellent alternatives to PageRank in estimating node prominence in bipartite social networks, primarily because each algorithm is able to holistically capture the topology of bipartite networks. In patient-provider networks, BiRank and CoHITS are better able to capture patients' preferences to seek opioids from simultaneous providers simultaneously and to recommend and seek the same sets of generous opioid providers. Moreover, our evidence suggests that estimating centrality on patient-provider networks weighted by the total potency of opioids prescribed (MME) by a provider may be able to improve our ability to identify potential fraudulent behavior because these estimates also capture patients' preferences for providers that prescribe potent drugs. Identifying individuals who are at risk of future opioid overdose is critical to the U.S. public health. We hope in earnest that these methods might be implemented to guide future prescription drug monitoring programs in order to reduce overdose mortality and improve access to addiction services.

Although we believe that more studies are necessary to determine whether bipartite centrality algorithms generalize well to other social contexts, we hope that our evidence will motivate future research to begin estimating bipartite network centrality with CoHITS and BiRank instead of PageRank. These algorithms are relatively easy to implement in modern statistical packages (Aronson & Yang 2020; Yang 2020; Yang et al. 2020), and could be particularly useful in future studies of patient or provider behavior using medical claims data. Even without further studies regarding their accuracy, we believe that CoHITS and BiRank's ability to estimate centrality directly on bipartite social networks offers a substantial theoretical advantage over PageRank. The peace of mind in knowing that CoHITS and BiRank do not require network distortion to operate might alone be enough to warrant their increased adoption in future studies of node prominence in bipartite social networks.

References

Ali MM, Dowd WN, Classen T, Mutter R, Novak SP. 2017. Prescription Drug Monitoring Programs, Nonmedical Use of Prescription Drugs, and Heroin Use: Evidence from the National Survey of Drug Use and Health. *Addictive Behaviors*. 69:65–77

Allison PD. 1984. *Event History Analysis: Regression for Longitudinal Event Data*. Sage

Aronson B, Yang K-C. 2020. *Birankr: Ranking Nodes in Bipartite and Weighted Networks*

Bahmani B, Chowdhury A, Goel A. 2010. Fast Incremental and Personalized PageRank. *Proceedings of the VLDB Endowment*. 4(3):173–84

Barnett ML, Olenski AR, Jena AB. 2017. Opioid-Prescribing Patterns of Emergency Physicians and Risk of Long-Term Use. *New England Journal of Medicine*. 376(7):663–73

Bonacich P. 1991. Simultaneous Group and Individual Centralities. *Social Networks*. 13(2):155–68

Borgatti SP, Everett MG. 1997. Network Analysis of 2-mode Data. *Social Networks*. 19(3):243–269

Boucher V, Fortin B. 2016. Some Challenges in the Empirics of the Effects of Networks. *The Oxford Handbook on the Economics of Networks*. 277–302

Cepeda MS, Fife D, Chow W, Mastrogiovanni G, Henderson SC. 2012. *Assessing Opioid Shopping Behaviour*. *Drug Safety*. 35(4):325–334

Compton WM, Boyle M, Wargo E. 2015. Prescription Opioid Abuse: Problems and Responses. *Preventive Medicine*. 80:5–9

Compton WM, Jones CM, Baldwin GT. 2016. Relationship between Nonmedical Prescription-opioid use and heroin use. *New England Journal of Medicine*. 374(2):154–163

Cox DR, Oakes D. 1984. *Analysis of Survival Data*, Vol. 21. CRC Press

Deng H, Lyu MR, King I. 2009. A Generalized Co-HITS Algorithm and Its Application to Bipartite Graphs. *Proceedings of the 15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining - KDD '09*, p. 239. Paris, France: ACM Press

Devi P, Gupta A, Dixit A. 2014. Comparative Study of HITS and PageRank Link Based Ranking Algorithms. *International Journal of Advanced Research in Computer and Communication Engineering*. 3(2):5749–5754

Everett MG. 2016. Centrality and the Dual-Projection Approach for Two-Mode Social Network Data. *Methodological Innovations*. 9:205979911663066

Everett MG, Borgatti SP. 2013. The Dual-Projection Approach for Two-Mode Networks. *Social Networks*. 35(2):204–10

Friedkin NE, Johnsen EC. 2014. Two Steps to Obfuscation. *Social Networks*. 39:12–13

Fujimoto K, Chou C-P, Valente TW. 2011. The Network Autocorrelation Model Using Two-Mode Data: Affiliation Exposure and Potential Bias in the Autocorrelation Parameter. *Social Networks*. 33(3):231–43

Grassi R, Calderoni F, Bianchi M, Torriero A. 2019. Betweenness to Assess Leaders in Criminal Networks: New Evidence Using the Dual Projection Approach. *Social Networks*. 56:23–32

Guillaume J-L, Latapy M. 2004. Bipartite Structure of all Complex Networks. *Information Processing Letters*. 90(5):215–221

Han B, Compton WM, Jones CM, Cai R. 2015. Nonmedical Prescription Opioid Use and Use Disorders Among Adults Aged 18 Through 64 Years in the United States, 2003-2013. *JAMA*. 314(14):1468

Hanneman RA, Riddle M. 2005. *Introduction to Social Network Methods*

He X, Gao M, Kan M-Y, Wang D. 2017. BiRank: Towards Ranking on Bipartite Graphs. *IEEE Transactions on Knowledge and Data Engineering.* 29(1):57–71

Hedegaard H, Miniño AM, Warner M. 2020. Drug Overdose Deaths in the United States, 1999-2018

Kleinberg JM. 1999. Authoritative Sources in a Hyperlinked Environment. *J. ACM*. 46(5):604–632

Latapy M, Magnien C, Vecchio ND. 2008. Basic Notions for the Analysis of Large Two-Mode Networks. *Social Networks*. 30(1):31–48

Lilly JD. 2019. The Failure of Prescription Drug Monitoring Programs. *Journal of American Physicians and Surgeons*. 24(2), 56-59.

Martyres RF, Clode D, Burns JM. 2004. Seeking Drugs or Seeking Help? Escalating "Doctor Shopping" by Young Heroin Users before Fatal Overdose. *Medical Journal of Australia*. 180(5):211–14

Maxwell JC. 2011. The Prescription Drug Epidemic in the United States: A Perfect Storm. *Drug and Alcohol Review*. 30(3):264–270

McDonald DC, Carlson KE. 2013. Estimating the Prevalence of Opioid Diversion by "Doctor Shoppers" in the United States. *PLoS ONE*. 8(7):e69241

McDonald DC, Carlson KE. 2014. The Ecology of Prescription Opioid Abuse in the USA: Geographic Variation in Patients' Use of Multiple Prescribers ("doctor shopping"): Geography of Doctor Shopping for Opioids. *Pharmacoepidemiol Drug Safety*. 23(12):1258–67

Melamed D. 2014. Community Structures in Bipartite Networks: A Dual-Projection Approach. *PLoS ONE*. 9(5):e97823

Moody J, Light R. 2006. A View from Above: The Evolving Sociological Landscape. *The American Sociologist*. 37(2):67–86

Morris BJ, Zumsteg JW, Archer KR, Cash B, Mir HR. 2014. Narcotic Use and Postoperative Doctor Shopping in the Orthopaedic Trauma Population. *The Journal of Bone & Joint Surgery*. 96(15):1257–1262

Mouw T, Entwisle B. 2006. Residential Segregation and Interracial Friendship in Schools. *American Journal of Sociology*. 112(2):394–441

Newman ME. 2001. Scientific Collaboration Networks. II. Shortest Paths, Weighted Networks, and Centrality. *Physical Review E*. 64(1):016132

Newman ME, Strogatz SH, Watts DJ. 2001. Random Graphs with Arbitrary Degree Distributions and their Applications. *Physical Review E*. 64(2):026118

Nordmann S, Pradel V, Lapeyre-Mestre M, Frauger E, Pauly V, Thirion X, Mallaret M, Jouanjus, E, Micallef, J. Doctor Shopping Reveals Geographical Variations in Opioid Abuse. *Pain Physician*. 13

Ong M-S, Olson KL, Cami A, Liu C, Tian F, Selvam N, Mandl KD. 2016. Provider Patient-Sharing Networks and Multiple-Provider Prescribing of Benzodiazepines. *Journal of General Internal Medicine*. 31(2):164–71

Otte E, Rousseau R. 2002. Social Network Analysis: A Powerful Strategy, Also for the Information Sciences. *Journal of Information Science*. 28(6):441–453

Page L, Brin S, Motwani R, Winograd T. 1999. The PageRank Citation Ranking: Bringing Order to the Web. Stanford InfoLab

Peirce GL, Smith MJ, Abate MA, Halverson J. 2012. Doctor and Pharmacy Shopping for Controlled Substances. *Medical Care*. 50:494–500

Perry B, Freeman P, Oser C, Martel M, Talbert J. 2015. Can Social Network Analysis be Used to Identify Doctor Shoppers? *Value in Health*. 18(3):A128

Perry BL, Odabas M, Yang K-C, Lee B, Kaminski P, Aronson B, Ahn Y-Y, Oser CB, Freeman PR, Talbert JC. 2020. New Means New Measures: Assessing Traditional and Social Network Indicators of Prescription drug-seeking Behavior over Ten Years of the Opioid Epidemic

Perry BL, Yang KC, Kaminski P, Odabas M, Park J, Martel M, Oser CB, Freeman PR, Ahn Y-Y, Talbert J. 2019. Co-Prescription Network Reveals Social Dynamics of Opioid Doctor Shopping. *PloS one*. 14(10):

Pradel V, Frauger E, Thirion X, Ronfle E, Lapierre V, Masut A, Coudert C, Blin O, Micallef J. 2009. Impact of a Prescription Monitoring Program on Doctor-Shopping for High Dosage Buprenorphine. *Pharmacoepidemiological Drug Safety*. 18:36–43

Rui X, Li M, Li Z, Ma W-Y, Yu N. 2007. Bipartite Graph Reinforcement Model for Web Image Annotation. *Proceedings of the 15th ACM International Conference on Multimedia*, pp. 585–594

Sakamoto Y, Ishiguro M, Kitagawa G. 1986. Akaike Information Criterion Statistics. *Dordrecht, The Netherlands: D. Reidel*. 81:

Sansone RA, Sansone LA. 2012. Doctor Shopping: A Phenomenon of Many Themes. *Innovations in Clinical Neuroscience*. 9(11–12):42

Smith JA, Moody J. 2013. Structural Effects of Network Sampling Coverage I: Nodes Missing at Random. *Social Networks*. 35(4):652–668

Stein BD, Mendelsohn J, Gordon AJ, Dick AW, Burns RM, Sorbero M, Shin RA, Liccardo Pacula R. 2017. Opioid Analgesic and Benzodiazepine Prescribing among Medicaid-

Enrollees with Opioid use Disorders: The Influence of Provider Communities. *Journal of Addictive Diseases*. 36(1):14–22

Takahashi Y, Ishizaki T, Nakayama T, Kawachi I. 2016. Social Network Analysis of Duplicative Prescriptions: One-Month Analysis of Medical Facilities in Japan. *Health Policy*. 120(3):334–41

Thomas CP, Fullerton CA, Kim M, Montejano L, Lyman DR, Dougherty RH, Daniels AS, Ghose SS, Delphin-Rittmon ME. 2014. Medication-Assisted Treatment with Buprenorphine: Assessing the Evidence. *Psychiatric Services*. 65(2):158–170

Uzzi B, Spiro J. 2005. Collaboration and Creativity: The Small World Problem. *American Journal of Sociology*. 111(2):447–504

Wang DJ, Shi X, McFarland DA, Leskovec J. 2012. Measurement Error in Network Data: A Re-classification. *Social Networks*. 34(4):396–409

Wasserman S, Faust K, others. 1994. *Social Network Analysis: Methods and Applications*, Vol. 8. Cambridge University Press

Wesson DR, Smith DE. 1990. Prescription Drug Abuse. Patient, Clinician, and Cultural Responsibilities. *Western Journal of Medicine*. 152:613–616

Wilsey BL, Fishman SM, Gilson AM, Casamalhuapa C, Baxi H, Zhang H, Li CS. 2010. Profiling Multiple Provider Prescribing of Opioids, Benzodiazepines, Stimulants, and Anorexics. *Drug and Alcohol Dependence*. 112(1–2):99–106

World Health Organization. 1978. International Classification of Diseases:[9th] Ninth Revision, Basic Tabulation List with Alphabetic Index

Yang K-C. 2020. *Birankpy: Ranking Nodes in Bipartite and Weighted Networks*

Yang K-C, Aronson B, Ahn Y-Y. 2020. *BiRank: Fast and Flexible Ranking on Bipartite Networks with R and Python*

Yang Z, Wilsey B, Bohm M, Weyrich M, Roy K, Ritley D, Jones C, Melnikow J. 2015. Defining Risk of Prescription Opioid Overdose: Pharmacy Shopping and Overlapping Prescriptions Among Long-Term Opioid Users in Medicaid. *The Journal of Pain*. 16(5):445–53

Zhou T, Ren J, Medo M, Zhang Y-C. 2007. Bipartite Network Projection and Personal Recommendation. *Physical Review E*. 76(4):046115

Sources of Finances

This research was funded by the National Institute on Drug Abuse (R01 DA039928), an Indiana University Addictions Grand Challenge Grant, and a University of Kentucky Interdisciplinary Research Collaborations Seed Grant.

Acknowledgments

The authors thank Byungkyu Lee, Patrick Kaminski, Hank Green, Wanying Zhao, and Sirag Erkol for their helpful discussions. We would also like to thank the information technology team at the Indiana University Network Science Institute for their technical support.